

Week 4 Workshop

Based on the code of week 3, using Pytorch and Numpy to apply the following regularisers.

Technique	Formula	Implementation Tip
L2	$\text{Loss} + \lambda \sum W^2$	PyTorch: <code>weight_decay</code> , NumPy: <code>+2λW</code> in gradient
L1	$\text{Loss} + \lambda \sum W $	PyTorch: manual penalty, NumPy: <code>+λ · sign(W)</code>
Dropout	Random zeros	Scale by <code>1/(1-p)</code> during training
Label Smoothing	Soft targets	Confidence: <code>1-ε</code> , others: <code>ε/(K-1)</code>
Data Aug	Transform inputs	Apply only during training